FX Trading Lab — research protocol report

Strategy 1.1 (strategy_v1_1.yaml) · EURUSD · 2025-01-01 → 2025-12-31 · generated 2026-11-06 08:27 UTC · EURUSD_2025.csv

Educational research tool. This software is an educational research and backtesting tool. It is not investment advice, not a trading signal service, and it has no broker connection or live execution. All results are simulated on historical data using the recorded strategy and execution assumptions. Simulated results are not evidence of a profitable edge and historical performance does not guarantee future results. The author is not a licensed or registered financial adviser. Trading leveraged foreign exchange involves a high risk of loss; nothing here should be used to place real trades.

Reproducibility

Field	Value
Dataset SHA-256	f3a0e1fdfdd06863017eb449a149db8e2dc05032f80a5ce5e8ccbf81030070ae
Configuration hash	9454655f074bee1463a4ec10d62ee672a3c1466ac0a974498bdc14ccf0c050b9
Execution model	first_executable_m1_price_after_bos
Spread sources	HISTORICAL_SPREAD, FIXED_CONFIGURED_SPREAD, NO_SPREAD_DATA
Slippage	0.1 pips
Timezone database	2026.5
Software	0.5.0 · source-sha256:17085b1bee7ce590c7fc836fc484ce6f8797bda3b7186b9fe8d01a154a58cbc9

Data

Rows	Gaps	Duplicates	Invalid	Weekend rows	Spread column	Source timezone
372024	965	0	0	6259	no	embedded_offsets

■ Data gaps detected; no candles were fabricated.
■ Weekend-dated bars detected; verify source sessions and market closures.

Research protocol

Fixed-rule baseline evaluated on independent periods, rolling out-of-sample folds, a bootstrap of the trade sequence, one-at-a-time parameter variations and random entries that pay the same costs. Only out-of-sample figures may be quoted, always next to the random-benchmark percentile and the sensitivity table.

Baseline

Period	Trades	Expectancy	Total	Win rate	PF	Max DD	Sharpe (ann.)
full sample	10	-0.10R	-0.96R	50 %	0.78	+1.89R	-0.35

Independent chronological periods

Period	Trades	Expectancy	Total	Win rate	PF	Max DD	Sharpe (ann.)
train	3	-0.57R	-1.72R	33 %	0.14	+1.72R	-1.51
validation	2	+0.62R	+1.24R	100 %	—	+0.00R	2.92
out_of_sample	5	-0.10R	-0.48R	40 %	0.80	+1.42R	-0.52

Out-of-sample metrics are the only ones that may be quoted as evidence; the baseline has no fitted parameters, so train/validation serve as independent replications.

Rolling out-of-sample (walk-forward)

9 folds · train 120 d, test 30 d. Concatenated OOS: 7 trades, +0.76R, expectancy +0.11R, 2/4 folds positive.

Period	Trades	Expectancy	Total	Win rate	PF	Max DD	Sharpe (ann.)
fold 1 OOS	0	+0.00R	+0.00R	0 %	—	+0.00R	—
fold 2 OOS	0	+0.00R	+0.00R	0 %	—	+0.00R	—
fold 3 OOS	0	+0.00R	+0.00R	0 %	—	+0.00R	—
fold 4 OOS	0	+0.00R	+0.00R	0 %	—	+0.00R	—
fold 5 OOS	0	+0.00R	+0.00R	0 %	—	+0.00R	—
fold 6 OOS	2	+0.62R	+1.24R	100 %	—	+0.00R	4.54
fold 7 OOS	1	-0.42R	-0.42R	0 %	0.00	+0.42R	-3.31
fold 8 OOS	3	+0.31R	+0.94R	67 %	1.94	+1.00R	1.83
fold 9 OOS	1	-1.00R	-1.00R	0 %	0.00	+1.00R	-9.17

Monte Carlo (trade-sequence bootstrap)

Paths	Trades / path	p(losing)	Final R p5	p50	p95	Max DD p50	Max DD p95
1000	10	62 %	-5.06R	-0.97R	+3.82R	+2.84R	+5.72R

Cost-matched random benchmark

Baseline expectancy -0.10R sits at percentile 46 of 200 simulations of random entries with identical stop distance, target, costs and session expiry (random median -0.05R/trade). Below the 95th percentile the result is indistinguishable from random timing.

Parameter sensitivity (one change at a time)

Parameter	Value	Trades	Expectancy	Total	Max DD
baseline	—	10	-0.10R	-0.96R	+1.89R
strategy:target.baseline	1.5R	10	-0.10R	-0.96R	+1.89R
strategy:target.baseline	2.0R	10	-0.10R	-0.96R	+1.89R
strategy:target.baseline	2.5R	10	-0.10R	-0.96R	+1.89R
strategy:target.baseline	3.0R	10	-0.10R	-0.96R	+1.89R
execution:slippage.baseline_pips	0.0	10	-0.08R	-0.84R	+1.82R

Parameter	Value	Trades	Expectancy	Total	Max DD
execution:slippage.baseline_pips	0.1	10	-0.10R	-0.96R	+1.89R
execution:slippage.baseline_pips	0.2	10	-0.11R	-1.07R	+1.96R
execution:slippage.baseline_pips	0.5	8	-0.22R	-1.75R	+2.50R
strategy:risk.baseline_equity_risk_fraction	0.0025	10	-0.10R	-0.96R	+1.89R
strategy:risk.baseline_equity_risk_fraction	0.005	10	-0.10R	-0.96R	+1.89R
strategy:risk.baseline_equity_risk_fraction	0.01	10	-0.10R	-0.96R	+1.89R
strategy:zones.expiry_m15_bars	4	10	-0.10R	-0.96R	+1.89R
strategy:zones.expiry_m15_bars	8	10	-0.10R	-0.96R	+1.89R
strategy:zones.expiry_m15_bars	12	10	-0.10R	-0.96R	+1.89R
strategy:setup.lifetime_m15_bars_from_sweep_confirmation	6	10	-0.13R	-1.28R	+1.89R
strategy:setup.lifetime_m15_bars_from_sweep_confirmation	8	10	-0.10R	-0.96R	+1.89R
strategy:setup.lifetime_m15_bars_from_sweep_confirmation	12	11	-0.05R	-0.58R	+1.51R
strategy:displacement.minimum_atr	0.75	9	-0.15R	-1.33R	+2.27R
strategy:displacement.minimum_atr	1.0	10	-0.10R	-0.96R	+1.89R
strategy:displacement.minimum_atr	1.25	11	-0.18R	-1.96R	+2.89R
strategy:stop.reference	sweep_extreme	12	-0.00R	-0.04R	+2.22R
strategy:stop.reference	zone_distal	10	-0.10R	-0.96R	+1.89R
strategy:stop.reference	leg1_origin	14	+0.44R	+6.13R	+2.00R
strategy:stop.max_distance_atr	3.0	10	-0.10R	-0.96R	+1.89R
strategy:stop.max_distance_atr	4.0	13	-0.05R	-0.71R	+2.45R
strategy:stop.max_distance_atr	5.0	15	+0.16R	+2.42R	+2.00R
strategy:liquidity.level_max_age_m15_bars	None	11	+0.32R	+3.52R	+1.72R
strategy:liquidity.level_max_age_m15_bars	960	10	-0.10R	-0.96R	+1.89R
strategy:liquidity.level_max_age_m15_bars	1920	10	-0.10R	-0.96R	+1.89R
strategy:liquidity.level_max_age_m15_bars	3840	11	+0.09R	+1.04R	+1.72R
multi:session	london_open_09:00-10:30	10	-0.10R	-0.96R	+1.89R
multi:session	new_york_open_15:00-16:30	10	-0.08R	-0.77R	+2.06R

One-at-a-time variations (and labelled multi-key session experiments); large swings between neighbouring values indicate parameter fragility. Experiments are not the baseline and must not be reported as such.

Limitations

- Simulated fills on historical M1 data; no live execution was involved.
- Costs follow config/execution.yaml; provider spreads are used when present.
- A positive result in one period is not evidence of a robust edge.
- Historical performance does not guarantee future results.

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